SENTINEL AI 2.0 + GEMMA FLOWPILOT

PROJECT IDEA

What is Sentinel AI?

Companies are rapidly adopting AI tools like Google Gemini and ChatGPT.

This creates two major enterprise challenges:

🔒 Sensitive Data Exposure

Sensitive company data, passwords, customer records, or financial reports may accidentally be shared with AI models.

⚠️ AI Jailbreaks & Hacks

Hackers may try to manipulate AI models using
and jailbreak payload attacks.

Sentinel AI sits between employees and the AI model, scanning every prompt before it reaches the AI.

👤 Employee

🛡️ Sentinel AI Gateway

🤖 Google Gemini / Gemma

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CORE SECURITY PRINCIPLE

Zero Trust Security Architecture

"Never Trust. Always Verify."

Every request is cryptographically verified before accessing the AI mod

Core Zero-Trust Features:

- Verify Every Single Prompt in Real Time
- Detect & Redact Sensitive PII / Passwords
- Detect Malicious Prompts & Code Injections
- Block Unsafe Requests Instantly (Risk >= 0.50)
- Allow Only Safe, Sanitized Prompts

HOLOGRAPHIC VERIF

User Prompt Pa

↓

[100Hz Cryptographic Ver

↓

✅ ALLOW
(Safe Prompt)

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IDENTITY & ACCESS

Secure Role-Based Authentication

JWT (JSON Web Token) securely verifies every user identity and session authorization.

Super Admin

Level 5 Access

Full System Control, Emergency Lockdown Mode, Cryptographic Re-keying.

Security Analyst

SecOps Audit

Real-time Telemetry Stream Inspection, Threat Analysis & Policy Review.

Standard

Standard User

Filtered AI Assist
Masking Enforce

KEY AUTHENTICATION BENEFITS:

✓ Secure Login | ✓ 256-Bit Session Authentication | ✓ Strict Role-Based Access Control (RBAC) | ✓ Prevents Unauthorized Admini

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FINANCIAL INTELLIGENCE

Gemma FlowPilot — AI Financial Copilot

Gemma FlowPilot is an AI Financial Copilot that empowers businesses to manage their cashflow and runway.

📊 30-Day Cashflow Forecast

Predicts liquid balance (₹85,000) vs net monthly burn rate (₹42,000/mo).

📉 Business Runway Prediction

Monitors solvency timeline, alerting managers

🎛️ AI What-If Simulation

Live sliders for payment delays (0-60d), supplier extensions, and staff hires.

✉️ Invoice Collection Bot

Automated payment reminder outreach via Wh

💡 Decision Support Engine

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MACRO FINANCIAL SCENARIOS

Scenario Simulation Engine

Simulates three glowing financial macroeconomic states on the 30-Day Runway Chart:

😊 Optimistic

Solvent & Growing

Customers pay invoices on time.
Cash balance rises to ₹1,55,000.
Business remains financially healthy.

😬 Crunch

Tight Liquidity

Some customers delay payments (14-30d).
Cashflow balance drops to ₹18,000.
Working capital requires monitoring.

🚨 Crisis

Insolvency Risk

Many invoices
Balance drops
Business receiv

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AUTOMATED CUSTOMER OUTREACH

AI Invoice Collection Bot

Gemma automatically reminds overdue customers through WhatsApp Business & Corporate Email:

Friendly Reminder

Gentle Nudge

"Hi Sree Fabrics, hope business is going great! Just a gentle check-in regarding invoice INV-002 (₹1,20,000)..."

Firm Reminder

5-Day Deadline

"Hello Kavya Boutique, invoice INV-001 (₹45,000) is 38 days overdue. Please arrange payment by Friday..."

Final No

48-Hour Warnin

"URGENT NOTIC overdue. Please action..."

Common AI Threats & Adversarial Attacks

Four major attack vectors targeting enterprise AI implementations:

1. Prompt Injection

Attempting to override system instructions and force the AI into unintended actions.

2. Jailbreak Attacks

Using roleplay tricks ("DAN - Do Anything Now"

3. Sensitive Data Extraction

Tricking the LLM into dumping secret system prompts, API keys, or user records.

4. Malicious Code Injection

Sending harmful scripts such as Remote Code E payloads.

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DEFENSE ENGINE

AI Security Pipeline Workflow

End-to-End Inspection Pipeline:

1. User Prompt

Input captured in UI

2. PII Detection

Client-side ZK Redactor

3. Jailbreak Detection

AST & Pattern Scanner

4. Risk Scoring

100Hz ML Risk Engine

✅ **Safe Prompt (Risk < 0.50)**

❌ **Unsafe Prompt (Risk >= 0.50)**

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10 FLAGSHIP FEATURES

Platform Features Overview

- 🛡️ Zero Trust Gateway
- 🔑 JWT Authentication
- 🙈 PII Masking Redactor
- 🚨 Prompt Injection Scanner
- 🛑 Jailbreak Interceptor
- 📈 100Hz Risk Scoring Engine
- 📊 30-Day Cashflow Forecast
- 🎛️ Scenario Simulation

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CYBER ARCHITECTURE

Technology Stack & Architecture

FRONTEND

React 19, Tailwind CSS, Vite

BACKEND

Node.js, Express.js REST API

DATABASE

MongoDB / In-Memory Zero-Trust Store

AI MODELS

Google Gemini 1.5, Gemma 2B/7B LoR

SECURITY LAYER

JWT Bearer, Zero Trust, PII AST Scanner

ANALYTICS & CHARTS

Recharts Responsive Data Engine

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INDUSTRY USE CASES

Real-World Enterprise Applications

- 🏦 Banks & Financial Services
- 🏥 Healthcare Providers
- 🛡️ Insurance Companies
- 🏛️ Government Agencies
- 💻 IT & Tech Enterprise
- 🎓 Universities & Higher Ed

Why Sentinel AI? (Comparison Matrix)

Feature Criteria	Traditional AI Tools	Sentinel AI 2.0
Core Capability	Answers Questions Only	Answers + Protects E
Zero Trust Gateway	❌ No Zero Trust Gateway	✅ 100Hz Zero Trust
PII Protection	❌ No PII Redaction	✅ Automatic ZK PII
Attack Detection	❌ No Jailbreak Detection	✅ 0.96 Critical Jailb
Financial Copilot	❌ No Financial Intelligence	✅ Gemma 30-Day F

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STRATEGIC VISION

Future Roadmap & Expansion

🎙️ AI Voice Security & Audio Gateway

📄 OCR Document Scanner & PII Sanitizer

📱 Native iO

🌐 Multi-Language Zero-Trust Support

🤖 Multi-LLM Gateway (ChatGPT + Claude + Gemini)

📊 Enterpris

✉️ AI Phishing & Malware Detection

☁️ Enterprise One-Click Cloud Deployment

✨ Predictive